

Homework 2 VCF Lab Practice

0 README

文档中VCF文件的第34行为

```
gi|9626243|ref|NC_001416.1| 1104 . C A 225 .  
DP=43;VDB=0.162843;SGB=-0.693079;MQSB=0.981133;MQOF=0;AC=2;AN=2;DP4=0,0,8,21;MQ=41 GT:PL 1/1:255,84,0
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而我的VCF文件的第35行为

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gi|9626243|ref|NC_001416.1| 1104 . C A 228.378 .  
DP=43;VDB=0.337802;SGB=-0.693141;RPBZ=-1.32279;MQBZ=-0.164399;MQSBZ=-0.638285;BQBZ=1.04957;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=1,0,10,27;MQ=41 GT:PL:AD  
1/1:255,96,0:1,36
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不仅细节不同，我的还有更多的数据

而我的第34行相差更大，怀疑是库的版本问题

1 按 INDEL 和 SNP 把突变条目区分开

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grep 'INDEL' tmp.vcf | grep -v '#' # INDEL 条目
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结果为

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gi|9626243|ref|NC_001416.1| 245 . ATT AT 216.329 .  
INDEL;IDV=33;IMF=0.942857;DP=35;VDB=0.537812;SGB=-0.69311;RPBZ=-1.49293;MQBZ=1.58942;MQSBZ=2.47759;BQBZ=-1.57202;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=1,0,16,15;MQ=36  
GT:PL:AD 1/1:243,79,0:1,31  
gi|9626243|ref|NC_001416.1| 353 . GCTGAAATTGA G 222.98 .  
INDEL;IDV=24;IMF=0.685714;DP=35;VDB=0.00608521;SGB=-0.692831;RPBZ=1.47495;MQBZ=-2.92683;MQSBZ=1.11022;BQBZ=-1.80962;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=5,5,12,12;MQ=28  
GT:PL:AD 1/1:250,9,0:10,24  
gi|9626243|ref|NC_001416.1| 363 . A ATG 35.1841 .  
INDEL;IDV=8;IMF=0.228571;DP=35;VDB=0.00843033;SGB=-0.636426;RPBZ=-3.32157;MQBZ=2.68634;MQSBZ=-2.15744;BQBZ=0.587995;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=7,13,4,3;MQ=28  
GT:PL:AD 1/1:63,6,0:20,7  
gi|9626243|ref|NC_001416.1| 2817 . GA G 228.24 .  
INDEL;IDV=41;IMF=0.911111;DP=45;VDB=0.27095;SGB=-0.693144;RPBZ=-0.797953;MQBZ=-1.17465;MQSBZ=1.36281;BQBZ=-0.609359;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=2,1,20,19;MQ=41  
GT:PL:AD 1/1:255,80,0:3,39  
gi|9626243|ref|NC_001416.1| 2952 . CCCACC CCC 228.295 .  
INDEL;IDV=28;IMF=0.903226;DP=31;VDB=0.17228;SGB=-0.692914;RPBZ=-2.27241;MQBZ=-0.907724;MQSBZ=-0.823524;BQBZ=-0.0964065;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=0,3,13,12;MQ=39  
GT:PL:AD 1/1:255,47,0:3,25  
gi|9626243|ref|NC_001416.1| 3264 . CA C 212.288 .  
INDEL;IDV=41;IMF=0.803922;DP=51;VDB=0.063464;SGB=-0.693145;RPBZ=1.97114;MQBZ=-1.39289;MQSBZ=0.302321;BQBZ=0.842334;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=2,2,18,23;MQ=4
```

1 GT:PL:AD 1/1:239,84,0:4,41
 gi|9626243|ref|NC_001416.1| 3634 . ACGC AC 228.256 .
 INDEL;IDV=25;IMF=0.892857;DP=28;VDB=0.303244;SGB=-0.692914;RPBZ=-0.743494;MQBZ=-0.622649;MQSBZ=-0.475595;BQBZ=-1.2814;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,1,17,8;MQ=4
 1 GT:PL:AD 1/1:255,50,0:2,25
 gi|9626243|ref|NC_001416.1| 6290 . GT G 224.406 .
 INDEL;IDV=38;IMF=0.926829;DP=41;VDB=0.711061;SGB=-0.693143;RPBZ=0.951687;MQBZ=-0.402422;MQSBZ=1.20156;BQBZ=-0.408155;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,1,23,15;MQ=41
 GT:PL:AD 1/1:251,101,0:1,38
 gi|9626243|ref|NC_001416.1| 6762 . TCC TC 202.417 .
 INDEL;IDV=41;IMF=0.931818;DP=44;VDB=0.0299756;SGB=-0.693145;RPBZ=2.81869;MQBZ=-0.479813;MQSBZ=-0.117954;BQBZ=-1.52651;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,0,26,15;MQ=41
 GT:PL:AD 1/1:232,123,0:0,41
 gi|9626243|ref|NC_001416.1| 7297 . CCGTGTGGCAGAGCTGAA C 125.416 .
 INDEL;IDV=12;IMF=0.631579;DP=19;VDB=0.506425;SGB=-0.691153;RPBZ=-1.52195;MQBZ=-0.653966;MQSBZ=0.331455;BQBZ=-1.59605;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,0,9,9;MQ=16
 GT:PL:AD 1/1:155,54,0:0,18
 gi|9626243|ref|NC_001416.1| 7342 . TGC T 156.324 .
 INDEL;IDV=25;IMF=0.925926;DP=27;VDB=0.00864413;SGB=-0.651104;RPBZ=2.31632;MQBZ=-1.75226;MQSBZ=-2.02048;BQBZ=-1.12809;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,1,3,5;MQ=35
 GT:PL:AD 1/1:183,17,0:1,8
 gi|9626243|ref|NC_001416.1| 9964 . AGCAGG AG 228.218 .
 INDEL;IDV=24;IMF=0.774194;DP=31;VDB=0.016535;SGB=-0.692717;RPBZ=-2.27076;MQBZ=0.277667;MQSBZ=-1.68249;BQBZ=-0.361787;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=3,2,10,13;MQ=39
 GT:PL:AD 1/1:255,25,0:5,23
 gi|9626243|ref|NC_001416.1| 12411 . CAGCT C 228.374 .
 INDEL;IDV=33;IMF=0.916667;DP=36;VDB=0.0962593;SGB=-0.693127;RPBZ=0.944948;MQBZ=1.03187;MQSBZ=1.56101;BQBZ=-1.86305;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=2,1,15,18;MQ=40
 GT:PL:AD 1/1:255,70,0:3,33
 gi|9626243|ref|NC_001416.1| 12580 . TCACCC TC 228.19 .
 INDEL;IDV=31;IMF=0.815789;DP=38;VDB=0.247592;SGB=-0.693111;RPBZ=-2.91894;MQBZ=-0.679137;MQSBZ=-1.89611;BQBZ=-1.13878;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=2,3,15,16;MQ=39
 GT:PL:AD 1/1:255,38,0:5,31
 gi|9626243|ref|NC_001416.1| 13057 . AG A 207.379 .
 INDEL;IDV=36;IMF=0.878049;DP=41;VDB=0.126784;SGB=-0.693136;RPBZ=2.19316;MQBZ=-0.877223;MQSBZ=1.06738;BQBZ=-0.895873;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,1,14,21;MQ=41
 GT:PL:AD 1/1:234,83,0:2,35
 gi|9626243|ref|NC_001416.1| 13885 . TCCACGCC TCC 228.15 .
 INDEL;IDV=20;IMF=0.689655;DP=29;VDB=0.103823;SGB=-0.692067;RPBZ=-0.330105;MQBZ=-1.48682;MQSBZ=-0.269943;BQBZ=-0.931122;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=2,1,7,13;MQ=38
 GT:PL:AD 1/1:255,19,0:3,20
 gi|9626243|ref|NC_001416.1| 14556 . AGGATG AG 228.343 .
 INDEL;IDV=38;IMF=0.95;DP=40;VDB=0.277387;SGB=-0.693141;RPBZ=-2.35906;MQBZ=-1.78254;MQSBZ=1.47697;BQBZ=-0.908951;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,2,18,19;MQ=40
 GT:PL:AD 1/1:255,86,0:2,37
 gi|9626243|ref|NC_001416.1| 14917 . CCGGCG CCG 228.409 .
 INDEL;IDV=35;IMF=0.921053;DP=38;VDB=0.3497;SGB=-0.693136;RPBZ=1.29948;MQBZ=2.13501;MQSBZ=0.774597;BQBZ=-1.12422;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,0,21,14;MQ=40
 GT:PL:AD 1/1:255,93,0:1,35
 gi|9626243|ref|NC_001416.1| 15082 . CA C 202.363 .
 INDEL;IDV=43;IMF=0.877551;DP=49;VDB=0.432542;SGB=-0.693146;RPBZ=-0.838871;MQBZ=-1.14318;MQSBZ=-3.11415;BQBZ=-1.33488;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,2,27,16;MQ=4

1 GT:PL:AD 1/1:229,106,0:2,43
gi|9626243|ref|NC_001416.1| 17658 . GA G 228.322 .
INDEL;IDV=44;IMF=0.93617;DP=47;VDB=0.414993;SGB=-0.693145;RPBZ=0.326509;MQBZ=-0.611046;MQSBZ=-0.63901;BQBZ=0.22239;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,2,24,17;MQ=41
GT:PL:AD 1/1:255,91,0:3,41
gi|9626243|ref|NC_001416.1| 19073 . GTGA G 228.358 .
INDEL;IDV=50;IMF=0.892857;DP=56;VDB=0.212885;SGB=-0.693147;RPBZ=-1.93428;MQBZ=-0.633879;MQSBZ=-0.186151;BQBZ=-1.72175;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,3,33,16;MQ=40
GT:PL:AD 1/1:255,105,0:4,49
gi|9626243|ref|NC_001416.1| 20591 . AGAGG AG 228.389 .
INDEL;IDV=55;IMF=0.916667;DP=60;VDB=0.123734;SGB=-0.693147;RPBZ=-0.187298;MQBZ=0.51719;MQSBZ=-1.45686;BQBZ=-0.72314;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=2,0,26,27;MQ=41
GT:PL:AD 1/1:255,134,0:2,53
gi|9626243|ref|NC_001416.1| 21054 . AACC A 228.318 .
INDEL;IDV=47;IMF=0.854545;DP=55;VDB=0.0751605;SGB=-0.693147;RPBZ=0.131374;MQBZ=0.746048;MQSBZ=0.302249;BQBZ=-1.05755;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,3,24,23;MQ=40
GT:PL:AD 1/1:255,97,0:4,47
gi|9626243|ref|NC_001416.1| 21533 . AG A 228.332 .
INDEL;IDV=40;IMF=0.952381;DP=42;VDB=0.0289263;SGB=-0.693145;RPBZ=-2.36336;MQBZ=-0.223607;MQSBZ=1.08465;BQBZ=-0.975412;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=2,0,17,23;MQ=41
GT:PL:AD 1/1:255,95,0:2,40
gi|9626243|ref|NC_001416.1| 21972 . AATGGCAT A 228.266 .
INDEL;IDV=34;IMF=0.871795;DP=39;VDB=0.00490472;SGB=-0.693132;RPBZ=-0.189191;MQBZ=0.58888;MQSBZ=-1.36883;BQBZ=-0.839881;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=3,1,19,15;MQ=36
GT:PL:AD 1/1:255,59,0:4,34
gi|9626243|ref|NC_001416.1| 22454 . CCTC CC 228.393 .
INDEL;IDV=27;IMF=0.931035;DP=29;VDB=0.264618;SGB=-0.692562;RPBZ=-1.89393;MQBZ=-1.44516;MQSBZ=-1.13748;BQBZ=-1.65995;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,0,9,13;MQ=40
GT:PL:AD 1/1:255,56,0:1,22
gi|9626243|ref|NC_001416.1| 22473 . GAA G 210.351 .
INDEL;IDV=25;IMF=0.892857;DP=28;VDB=0.315921;SGB=-0.692067;RPBZ=-0.928731;MQBZ=-0.0411597;MQSBZ=-0.848734;BQBZ=-1.66763;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,1,7,13;MQ=40
GT:PL:AD 1/1:237,43,0:2,20
gi|9626243|ref|NC_001416.1| 23112 . TAAAAAA TAAAA 225.417 .
INDEL;IDV=34;IMF=0.944444;DP=36;VDB=0.0452312;SGB=-0.693132;RPBZ=0.0691192;MQBZ=-0.5073;MQSBZ=-1.4276;BQBZ=-1.34761;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,0,15,19;MQ=41
GT:PL:AD 1/1:255,102,0:0,34
gi|9626243|ref|NC_001416.1| 23386 . TTAC T 225.417 .
INDEL;IDV=30;IMF=0.909091;DP=33;VDB=0.163818;SGB=-0.693097;RPBZ=2.81919;MQBZ=1.92558;MQSBZ=-0.170697;BQBZ=-1.75664;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,0,17,13;MQ=41
GT:PL:AD 1/1:255,90,0:0,30
gi|9626243|ref|NC_001416.1| 25184 . GTT GT 228.371 .
INDEL;IDV=37;IMF=0.948718;DP=39;VDB=0.556018;SGB=-0.693136;RPBZ=-2.22918;MQBZ=-0.232495;MQSBZ=0.816497;BQBZ=-0.249101;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,0,19,16;MQ=42
GT:PL:AD 1/1:255,93,0:1,35
gi|9626243|ref|NC_001416.1| 25920 . ATTT ATT 222.388 .
INDEL;IDV=38;IMF=0.95;DP=40;VDB=0.672712;SGB=-0.693141;RPBZ=2.17312;MQBZ=2.31176;MQSBZ=0.220246;BQBZ=-0.631446;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,1,15,22;MQ=41
GT:PL:AD 1/1:249,99,0:1,37
gi|9626243|ref|NC_001416.1| 26294 . CTATTATT CTATT 228.396 .
INDEL;IDV=34;IMF=0.918919;DP=37;VDB=0.50051;SGB=-0.693132;RPBZ=-0.974015;MQBZ=-0.321934;MQSBZ=-0.41831;BQBZ=0;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,1,20,14;MQ=40

GT:PL:AD 1/1:255,80,0:2,34
gi|9626243|ref|NC_001416.1| 27234 . GCTCT GCT 225.417 .
INDEL;IDV=28;IMF=0.965517;DP=29;VDB=0.0904072;SGB=-0.693054;RPBZ=-1.67476;MQBZ=-0.339683;MQSBZ=1.54638;BQBZ=-1.59098;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,0,18,10;MQ=41
GT:PL:AD 1/1:255,84,0:0,28
gi|9626243|ref|NC_001416.1| 27472 . GCACACA GCACA 228.35 .
INDEL;IDV=34;IMF=0.871795;DP=39;VDB=0.376907;SGB=-0.693132;RPBZ=-0.462467;MQBZ=0.94951;MQSBZ=-1.08616;BQBZ=-2.28301;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,3,15,19;MQ=41
GT:PL:AD 1/1:255,72,0:3,34
gi|9626243|ref|NC_001416.1| 30577 . CTTCTATTTCT CTTCT 228.276 .
INDEL;IDV=39;IMF=0.764706;DP=51;VDB=0.0164896;SGB=-0.693144;RPBZ=-2.55488;MQBZ=-2.32421;MQSBZ=-0.968456;BQBZ=-0.953821;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=3,4,18,21;MQ=39
GT:PL:AD 1/1:255,60,0:7,39
gi|9626243|ref|NC_001416.1| 33181 . AGGGGG AGGG 228.339 .
INDEL;IDV=39;IMF=0.866667;DP=45;VDB=0.169757;SGB=-0.693144;RPBZ=1.70368;MQBZ=-0.812658;MQSBZ=1.43486;BQBZ=0.578169;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=2,1,19,20;MQ=41
GT:PL:AD 1/1:255,85,0:3,39
gi|9626243|ref|NC_001416.1| 34442 . TA T 193.293 .
INDEL;IDV=33;IMF=0.891892;DP=37;VDB=0.219282;SGB=-0.693127;RPBZ=-2.66695;MQBZ=3.25507;MQSBZ=-1.0328;BQBZ=-0.504286;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,2,15,18;MQ=41
GT:PL:AD 1/1:220,70,0:3,33
gi|9626243|ref|NC_001416.1| 36449 . AT A 228.362 .
INDEL;IDV=38;IMF=0.926829;DP=41;VDB=0.0615883;SGB=-0.693143;RPBZ=0.951563;MQBZ=-0.661371;MQSBZ=0.0616041;BQBZ=-0.366355;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,2,19,19;MQ=41
GT:PL:AD 1/1:255,91,0:2,38
gi|9626243|ref|NC_001416.1| 36957 . GTTTTT GTTT 228.282 .
INDEL;IDV=29;IMF=0.828571;DP=35;VDB=0.00810286;SGB=-0.693079;RPBZ=-1.05097;MQBZ=0.661699;MQSBZ=1.67168;BQBZ=-0.128447;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,2,12,17;MQ=41
GT:PL:AD 1/1:255,50,0:3,29
gi|9626243|ref|NC_001416.1| 38381 . AAC A 228.381 .
INDEL;IDV=37;IMF=0.880952;DP=42;VDB=0.283513;SGB=-0.693141;RPBZ=-0.641107;MQBZ=-0.763685;MQSBZ=-1.36583;BQBZ=-0.754591;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=3,0,12,25;MQ=41
GT:PL:AD 1/1:255,80,0:3,37
gi|9626243|ref|NC_001416.1| 39006 . ACC AC 208.206 .
INDEL;IDV=30;IMF=0.833333;DP=36;VDB=0.308515;SGB=-0.693097;RPBZ=-2.27168;MQBZ=-0.185695;MQSBZ=-1.05529;BQBZ=0.402448;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=3,0,11,19;MQ=41
GT:PL:AD 1/1:235,57,0:3,30
gi|9626243|ref|NC_001416.1| 39900 . GTTGCAT GT 228.404 .
INDEL;IDV=34;IMF=0.944444;DP=36;VDB=0.0309817;SGB=-0.693132;RPBZ=0;MQBZ=-1.14462;MQSBZ=-0.959235;BQBZ=-0.437332;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,1,18,16;MQ=41
GT:PL:AD 1/1:255,90,0:1,34
gi|9626243|ref|NC_001416.1| 40831 . AT A 225.417 .
INDEL;IDV=28;IMF=1;DP=28;VDB=0.184255;SGB=-0.693054;MQSBZ=0;BQBZ=-1.2684;MQ0F=0;AC=2;AN=2;DP4=0,0,12,16;MQ=42
GT:PL:AD 1/1:255,84,0:0,28
gi|9626243|ref|NC_001416.1| 41193 . GTTGT G 228.308 .
INDEL;IDV=37;IMF=0.925;DP=40;VDB=0.00133078;SGB=-0.69168;RPBZ=0.796405;MQBZ=1.54685;MQSBZ=2.24984;BQBZ=-1.57129;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=2,0,10,9;MQ=33
GT:PL:AD 1/1:255,38,0:2,19
gi|9626243|ref|NC_001416.1| 41253 . CGCTCACCGTATTG C 136.176 .
INDEL;IDV=27;IMF=0.818182;DP=33;VDB=0.0263282;SGB=-0.692562;RPBZ=-0.793625;MQBZ=-1.78062;MQSBZ=0.649234;BQBZ=-1.23304;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,2,15,7;MQ=19
GT:PL:AD 1/1:163,39,0:3,22

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gi|9626243|ref|NC_001416.1| 41678 . CTT C 228.41 .
INDEL;IDV=50;IMF=0.980392;DP=51;VDB=0.684289;SGB=-0.693147;RPBZ=-1.5293;MQBZ=2.1
0489;MQSBZ=-0.0591312;BQBZ=-0.551736;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,0,20,30;MQ=
41 GT:PL:AD 1/1:255,136,0:1,50
gi|9626243|ref|NC_001416.1| 42691 . CCGT C 228.412 .
INDEL;IDV=34;IMF=0.971429;DP=35;VDB=0.074891;SGB=-0.693127;RPBZ=-1.68419;MQBZ=-1
.1024;MQSBZ=0.255476;BQBZ=-1.63615;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,1,11,22;MQ=39
GT:PL:AD 1/1:255,87,0:1,33
gi|9626243|ref|NC_001416.1| 42798 . GCACC GC 228.403 .
INDEL;IDV=44;IMF=0.897959;DP=49;VDB=0.0646112;SGB=-0.693145;RPBZ=-1.7512;MQBZ=-0
.394671;MQSBZ=-1.72785;BQBZ=-1.51562;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,2,21,20;MQ=4
0 GT:PL:AD 1/1:255,100,0:2,41
gi|9626243|ref|NC_001416.1| 44172 . TGCA T 228.367 .
INDEL;IDV=42;IMF=0.893617;DP=47;VDB=0.0680788;SGB=-0.693145;RPBZ=0.724904;MQBZ=
0.144255;MQSBZ=0.422803;BQBZ=-1.69283;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=2,2,21,20;MQ
=41 GT:PL:AD 1/1:255,84,0:4,41
gi|9626243|ref|NC_001416.1| 44335 . CGG CG 225.417 .
INDEL;IDV=42;IMF=0.893617;DP=47;VDB=0.84515;SGB=-0.693146;RPBZ=0.949142;MQBZ=-0.
40071;MQSBZ=-1.0774;BQBZ=1.00443;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,0,20,22;MQ=41
GT:PL:AD 1/1:255,126,0:0,42
gi|9626243|ref|NC_001416.1| 45329 . AACGAT A 228.298 .
INDEL;IDV=31;IMF=0.885714;DP=35;VDB=0.0538266;SGB=-0.69311;RPBZ=-1.53047;MQBZ=-1
.19372;MQSBZ=0.305503;BQBZ=-0.750114;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=2,1,13,18;MQ=
38 GT:PL:AD 1/1:255,59,0:3,31
gi|9626243|ref|NC_001416.1| 45558 . TAA T 228.358 .
INDEL;IDV=39;IMF=0.975;DP=40;VDB=0.445349;SGB=-0.693144;RPBZ=-1.68959;MQBZ=-0.76
4719;MQSBZ=-0.699548;BQBZ=1.26194;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,1,21,18;MQ=40
GT:PL:AD 1/1:255,100,0:1,39
gi|9626243|ref|NC_001416.1| 45887 . ATG A 228.28 .
INDEL;IDV=46;IMF=0.901961;DP=51;VDB=0.227666;SGB=-0.693147;RPBZ=-3.64327;MQBZ=-0
.850331;MQSBZ=0.39101;BQBZ=-1.41179;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=3,1,18,28;MQ=4
1 GT:PL:AD 1/1:255,88,0:4,46

```

```
grep -v -e 'INDEL' -v -e '#' tmp.vcf # SNP 条目
```

结果为

```

gi|9626243|ref|NC_001416.1| 355 . T C 9.88941 .
DP=10;VDB=0.631962;SGB=-0.616816;MQSBZ=-1.22474;MQ0F=0;AC=2;AN=2;DP4=0,0,5,1;M
Q=35 GT:PL:AD 1/1:39,15,0:0,5
gi|9626243|ref|NC_001416.1| 360 . T G 8.13869 .
DP=13;VDB=0.00637373;SGB=-0.636426;MQSBZ=-1.77482;MQ0F=0;AC=2;AN=2;DP4=0,0,4,3;
MQ=31 GT:PL:AD 1/1:37,21,0:0,7
gi|9626243|ref|NC_001416.1| 1104 . C A 228.378 .
DP=43;VDB=0.337802;SGB=-0.693141;RPBZ=-1.32279;MQBZ=-0.164399;MQSBZ=-0.638285;B
QBZ=1.04957;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,0,10,27;MQ=41 GT:PL:AD
1/1:255,96,0:1,36
gi|9626243|ref|NC_001416.1| 1344 . G T 222.417 .
DP=37;VDB=0.730004;SGB=-0.693097;MQSBZ=0.816497;MQ0F=0;AC=2;AN=2;DP4=0,0,18,12;
MQ=41 GT:PL:AD 1/1:252,87,0:0,29
gi|9626243|ref|NC_001416.1| 2143 . C G 225.417 .

```

DP=46;VDB=0.942003;SGB=-0.693141;MQSBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,0,16,21;MQ=42
 GT:PL:AD 1/1:255,108,0:0,36
 gi|9626243|ref|NC_001416.1| 3316 . T C 228.39 .
 DP=59;VDB=0.4619;SGB=-0.693146;RPBZ=1.00132;MQBZ=-0.353553;MQSBZ=1.35534;BQBZ
 =1.07902;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,1,23,21;MQ=41 GT:PL:AD
 1/1:255,116,0:1,43
 gi|9626243|ref|NC_001416.1| 3406 . G T 205.417 .
 DP=40;VDB=0.326039;SGB=-0.693132;MQSBZ=-1.97765;MQ0F=0;AC=2;AN=2;DP4=0,0,16,18;
 MQ=41 GT:PL:AD 1/1:235,93,0:0,31
 gi|9626243|ref|NC_001416.1| 5812 . A C 29.1757 .
 DP=13;VDB=0.682725;SGB=-0.616816;RPBZ=-1.51357;MQBZ=1.01835;MQSBZ=-0.720082;BQ
 BZ=1.00905;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,1,3,3;MQ=29 GT:PL:AD 1/1:56,12,0:1,6
 gi|9626243|ref|NC_001416.1| 7089 . A C 198.416 .
 DP=26;VDB=0.0845165;SGB=-0.692562;MQSBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,0,6,16;MQ=42
 GT:PL:AD 1/1:228,66,0:0,22
 gi|9626243|ref|NC_001416.1| 9632 . T A 107.421 .
 DP=17;VDB=0.928906;SGB=-0.689466;MQSBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,0,7,9;MQ=42
 GT:PL:AD 1/1:139,45,2:0,14
 gi|9626243|ref|NC_001416.1| 9642 . T A 182.416 .
 DP=20;VDB=0.999266;SGB=-0.69168;MQSBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,0,8,11;MQ=42
 GT:PL:AD 1/1:212,54,0:0,18
 gi|9626243|ref|NC_001416.1| 12512 . G A 225.417 .
 DP=46;VDB=0.760072;SGB=-0.693145;MQSBZ=1.25727;MQ0F=0;AC=2;AN=2;DP4=0,0,21,20;
 MQ=41 GT:PL:AD 1/1:255,120,0:0,40
 gi|9626243|ref|NC_001416.1| 14897 . G C 225.417 .
 DP=38;VDB=0.584439;SGB=-0.693097;MQSBZ=1.06451;MQ0F=0;AC=2;AN=2;DP4=0,0,18,12;
 MQ=41 GT:PL:AD 1/1:255,90,0:0,30
 gi|9626243|ref|NC_001416.1| 18018 . G T 225.422 .
 DP=48;VDB=0.88965;SGB=-0.693141;MQSBZ=-0.35968;MQ0F=0;AC=2;AN=2;DP4=0,0,23,14;M
 Q=41 GT:PL:AD 1/1:255,108,0:0,36
 gi|9626243|ref|NC_001416.1| 21086 . G A 225.417 .
 DP=48;VDB=0.0317498;SGB=-0.693144;MQSBZ=-1.08195;MQ0F=0;AC=2;AN=2;DP4=0,0,15,24
 ;MQ=40 GT:PL:AD 1/1:255,117,0:0,39
 gi|9626243|ref|NC_001416.1| 21193 . T G 227.371 .
 DP=47;VDB=0.0974666;SGB=-0.693139;RPBZ=-1.54573;MQBZ=-0.438986;MQSBZ=-2.42835;B
 QBZ=0.891873;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,0,16,20;MQ=41 GT:PL:AD
 1/1:254,96,0:1,36
 gi|9626243|ref|NC_001416.1| 21707 . G T 225.417 .
 DP=40;VDB=0.357307;SGB=-0.69312;MQSBZ=0.37549;MQ0F=0;AC=2;AN=2;DP4=0,0,18,14;M
 Q=40 GT:PL:AD 1/1:255,96,0:0,32
 gi|9626243|ref|NC_001416.1| 21818 . G A 225.417 .
 DP=40;VDB=0.109551;SGB=-0.693097;MQSBZ=2.23927;MQ0F=0;AC=2;AN=2;DP4=0,0,16,14;
 MQ=40 GT:PL:AD 1/1:255,90,0:0,30
 gi|9626243|ref|NC_001416.1| 22778 . A T 225.421 .
 DP=41;VDB=0.601916;SGB=-0.693132;MQSBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,0,15,19;MQ=42
 GT:PL:AD 1/1:255,99,0:0,33
 gi|9626243|ref|NC_001416.1| 24099 . G C 210.417 .
 DP=25;VDB=0.771565;SGB=-0.692352;MQSBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,0,13,8;MQ=42
 GT:PL:AD 1/1:240,63,0:0,21
 gi|9626243|ref|NC_001416.1| 25563 . A G 206.357 .
 DP=29;VDB=0.298808;SGB=-0.693021;RPBZ=1.54767;MQBZ=-0.19245;MQSBZ=1;BQBZ=0.74
 3596;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,1,14,13;MQ=41 GT:PL:AD 1/1:233,70,0:1,27

gi|9626243|ref|NC_001416.1| 26443 . A C 184.416 .
DP=36;VDB=0.922442;SGB=-0.693079;MQSBZ=-0.829764;MQOF=0;AC=2;AN=2;DP4=0,0,16,13
;MQ=41 GT:PL:AD 1/1:214,84,0:0,28

gi|9626243|ref|NC_001416.1| 28553 . A C 225.417 .
DP=41;VDB=0.716375;SGB=-0.693143;MQSBZ=0;MQOF=0;AC=2;AN=2;DP4=0,0,19,19;MQ=42
GT:PL:AD 1/1:255,114,0:0,38

gi|9626243|ref|NC_001416.1| 32012 . A C 225.417 .
DP=38;VDB=0.320963;SGB=-0.693132;MQSBZ=0;MQOF=0;AC=2;AN=2;DP4=0,0,19,15;MQ=42
GT:PL:AD 1/1:255,102,0:0,34

gi|9626243|ref|NC_001416.1| 35035 . A T 225.417 .
DP=41;VDB=0.459904;SGB=-0.693143;MQSBZ=-1.51127;MQOF=0;AC=2;AN=2;DP4=0,0,20,18;
MQ=41 GT:PL:AD 1/1:255,111,0:0,37

gi|9626243|ref|NC_001416.1| 35145 . T A 225.417 .
DP=47;VDB=0.92921;SGB=-0.693146;MQSBZ=-1.10024;MQOF=0;AC=2;AN=2;DP4=0,0,23,19;M
Q=41 GT:PL:AD 1/1:255,120,0:0,40

gi|9626243|ref|NC_001416.1| 37052 . T G 188.416 .
DP=32;VDB=0.549894;SGB=-0.693079;MQSBZ=-2.0292;MQOF=0;AC=2;AN=2;DP4=0,0,12,17;M
Q=41 GT:PL:AD 1/1:218,87,0:0,29

gi|9626243|ref|NC_001416.1| 37844 . A T 187.416 .
DP=30;VDB=0.92808;SGB=-0.692914;MQSBZ=0;MQOF=0;AC=2;AN=2;DP4=0,0,16,9;MQ=42
GT:PL:AD 1/1:217,72,0:0,24

gi|9626243|ref|NC_001416.1| 39640 . C G 225.417 .
DP=42;VDB=0.45699;SGB=-0.693136;MQSBZ=1.08972;MQOF=0;AC=2;AN=2;DP4=0,0,16,19;M
Q=41 GT:PL:AD 1/1:255,102,0:0,34

gi|9626243|ref|NC_001416.1| 40311 . T A 218.417 .
DP=29;VDB=0.40188;SGB=-0.693021;MQSBZ=0;MQOF=0;AC=2;AN=2;DP4=0,0,15,12;MQ=42
GT:PL:AD 1/1:248,81,0:0,27

gi|9626243|ref|NC_001416.1| 40739 . C A 225.417 .
DP=41;VDB=0.647532;SGB=-0.693141;MQSBZ=0;MQOF=0;AC=2;AN=2;DP4=0,0,13,24;MQ=42
GT:PL:AD 1/1:255,111,0:0,37

gi|9626243|ref|NC_001416.1| 41376 . A C 174.416 .
DP=38;VDB=0.728779;SGB=-0.693021;MQSBZ=-2.18462;MQOF=0;AC=2;AN=2;DP4=0,0,17,10;
MQ=32 GT:PL:AD 1/1:204,81,0:0,27

gi|9626243|ref|NC_001416.1| 43666 . A C 225.417 .
DP=44;VDB=0.679576;SGB=-0.693145;MQSBZ=-1.29593;MQOF=0;AC=2;AN=2;DP4=0,0,18,22;
MQ=41 GT:PL:AD 1/1:255,120,0:0,40

gi|9626243|ref|NC_001416.1| 45569 . A T 225.417 .
DP=42;VDB=0.574691;SGB=-0.693132;MQSBZ=-1.15742;MQOF=0;AC=2;AN=2;DP4=0,0,19,15;
MQ=40 GT:PL:AD 1/1:255,99,0:0,33

gi|9626243|ref|NC_001416.1| 46398 . C G 182.416 .
DP=33;VDB=0.208281;SGB=-0.69311;MQSBZ=-0.28562;MQOF=0;AC=2;AN=2;DP4=0,0,19,12;M
Q=41 GT:PL:AD 1/1:212,93,0:0,31

gi|9626243|ref|NC_001416.1| 46744 . T A 214.421 .
DP=41;VDB=0.869978;SGB=-0.693132;MQSBZ=-0.691564;MQOF=0;AC=2;AN=2;DP4=0,0,11,23
;MQ=41 GT:PL:AD 1/1:244,90,0:0,30

gi|9626243|ref|NC_001416.1| 47808 . C G 225.417 .
DP=47;VDB=0.474038;SGB=-0.693143;MQSBZ=-0.899735;MQOF=0;AC=2;AN=2;DP4=0,0,17,21
;MQ=41 GT:PL:AD 1/1:255,114,0:0,38

2 分别统计 INDEL 和 SNP 条目的平均测序深度

```
cat tmp.vcf | grep -v "^#" | grep 'INDEL' | cut -f 8 | cut -d ";" -f 4 | awk  
'{print substr($0, 4)}' | awk '{sum += $0} END {print sum/NR}'  
# INDEL 条目
```

结果为

39.2642

```
cat tmp.vcf | grep -v "^#" | grep -v 'INDEL' | cut -f 8 | cut -d ";" -f 1 | awk  
'{print substr($0, 4)}' | awk '{sum += $0} END {print sum/NR}'  
# SNP 条目
```

结果为

36.4324

3 把 INDEL 条目再区分成 insertion 和 deletion 情况

```
grep -v '^#' tmp.vcf | grep 'INDEL' | awk '{if ($4 > $5) print $0}'  
# Deletion, 电脑屏幕太小, 不行折行根本复制不全
```

结果为

```
gi|9626243|ref|NC_001416.1| 245 . ATT AT 216.329 .  
INDEL;IDV=33;IMF=0.942857;DP=35;VDB=0.537812;SGB=-0.69311;RPBZ=-1.49293;MQBZ=1.5  
8942;MQSBZ=2.47759;BQBZ=-1.57202;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=1,0,16,15;MQ=36  
GT:PL:AD 1/1:243,79,0:1,31  
gi|9626243|ref|NC_001416.1| 353 . GCTGAAATTGA G 222.98 .  
INDEL;IDV=24;IMF=0.685714;DP=35;VDB=0.00608521;SGB=-0.692831;RPBZ=1.47495;MQBZ=  
-2.92683;MQSBZ=1.11022;BQBZ=-1.80962;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=5,5,12,12;MQ=2  
8 GT:PL:AD 1/1:250,9,0:10,24  
gi|9626243|ref|NC_001416.1| 2817 . GA G 228.24 .  
INDEL;IDV=41;IMF=0.911111;DP=45;VDB=0.27095;SGB=-0.693144;RPBZ=-0.797953;MQBZ=-1  
.17465;MQSBZ=1.36281;BQBZ=-0.609359;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=2,1,20,19;MQ=4  
1 GT:PL:AD 1/1:255,80,0:3,39  
gi|9626243|ref|NC_001416.1| 2952 . CCCACC CCC 228.295 .  
INDEL;IDV=28;IMF=0.903226;DP=31;VDB=0.17228;SGB=-0.692914;RPBZ=-2.27241;MQBZ=-0.  
907724;MQSBZ=-0.823524;BQBZ=-0.0964065;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=0,3,13,12;M  
Q=39 GT:PL:AD 1/1:255,47,0:3,25  
gi|9626243|ref|NC_001416.1| 3264 . CA C 212.288 .  
INDEL;IDV=41;IMF=0.803922;DP=51;VDB=0.063464;SGB=-0.693145;RPBZ=1.97114;MQBZ=-1.  
39289;MQSBZ=0.302321;BQBZ=0.842334;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=2,2,18,23;MQ=4  
1 GT:PL:AD 1/1:239,84,0:4,41  
gi|9626243|ref|NC_001416.1| 3634 . ACGC AC 228.256 .  
INDEL;IDV=25;IMF=0.892857;DP=28;VDB=0.303244;SGB=-0.692914;RPBZ=-0.743494;MQBZ=-  
0.622649;MQSBZ=-0.475595;BQBZ=-1.2814;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=1,1,17,8;MQ=4  
1 GT:PL:AD 1/1:255,50,0:2,25  
gi|9626243|ref|NC_001416.1| 6290 . GT G 224.406 .
```


INDEL;IDV=38;IMF=0.926829;DP=41;VDB=0.711061;SGB=-0.693143;RPBZ=0.951687;MQBZ=-0.402422;MQSBZ=1.20156;BQBZ=-0.408155;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,1,23,15;MQ=41 GT:PL:AD 1/1:251,101,0:1,38
gi|9626243|ref|NC_001416.1| 6762 . TCC TC 202.417 .
INDEL;IDV=41;IMF=0.931818;DP=44;VDB=0.0299756;SGB=-0.693145;RPBZ=2.81869;MQBZ=-0.479813;MQSBZ=-0.117954;BQBZ=-1.52651;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,0,26,15;MQ=41 GT:PL:AD 1/1:232,123,0:0,41
gi|9626243|ref|NC_001416.1| 7297 . CCGTGTGGCAGAGCTGAA C 125.416 .
INDEL;IDV=12;IMF=0.631579;DP=19;VDB=0.506425;SGB=-0.691153;RPBZ=-1.52195;MQBZ=-0.653966;MQSBZ=0.331455;BQBZ=-1.59605;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,0,9,9;MQ=16 GT:PL:AD 1/1:155,54,0:0,18
gi|9626243|ref|NC_001416.1| 7342 . TGC T 156.324 .
INDEL;IDV=25;IMF=0.925926;DP=27;VDB=0.00864413;SGB=-0.651104;RPBZ=2.31632;MQBZ=-1.75226;MQSBZ=-2.02048;BQBZ=-1.12809;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,1,3,5;MQ=35 GT:PL:AD 1/1:183,17,0:1,8
gi|9626243|ref|NC_001416.1| 9964 . AGCAGG AG 228.218 .
INDEL;IDV=24;IMF=0.774194;DP=31;VDB=0.016535;SGB=-0.692717;RPBZ=-2.27076;MQBZ=0.277667;MQSBZ=-1.68249;BQBZ=-0.361787;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=3,2,10,13;MQ=39 GT:PL:AD 1/1:255,25,0:5,23
gi|9626243|ref|NC_001416.1| 12411 . CAGCT C 228.374 .
INDEL;IDV=33;IMF=0.916667;DP=36;VDB=0.0962593;SGB=-0.693127;RPBZ=0.944948;MQBZ=1.03187;MQSBZ=1.56101;BQBZ=-1.86305;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=2,1,15,18;MQ=40 GT:PL:AD 1/1:255,70,0:3,33
gi|9626243|ref|NC_001416.1| 12580 . TCACCC TC 228.19 .
INDEL;IDV=31;IMF=0.815789;DP=38;VDB=0.247592;SGB=-0.69311;RPBZ=-2.91894;MQBZ=-0.679137;MQSBZ=-1.89611;BQBZ=-1.13878;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=2,3,15,16;MQ=39 GT:PL:AD 1/1:255,38,0:5,31
gi|9626243|ref|NC_001416.1| 13057 . AG A 207.379 .
INDEL;IDV=36;IMF=0.878049;DP=41;VDB=0.126784;SGB=-0.693136;RPBZ=2.19316;MQBZ=-0.877223;MQSBZ=1.06738;BQBZ=-0.895873;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,1,14,21;MQ=41 GT:PL:AD 1/1:234,83,0:2,35
gi|9626243|ref|NC_001416.1| 13885 . TCCACGCC TCC 228.15 .
INDEL;IDV=20;IMF=0.689655;DP=29;VDB=0.103823;SGB=-0.692067;RPBZ=-0.330105;MQBZ=-1.48682;MQSBZ=-0.269943;BQBZ=-0.931122;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=2,1,7,13;MQ=38 GT:PL:AD 1/1:255,19,0:3,20
gi|9626243|ref|NC_001416.1| 14556 . AGGATG AG 228.343 .
INDEL;IDV=38;IMF=0.95;DP=40;VDB=0.277387;SGB=-0.693141;RPBZ=-2.35906;MQBZ=-1.78254;MQSBZ=1.47697;BQBZ=-0.908951;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,2,18,19;MQ=40 GT:PL:AD 1/1:255,86,0:2,37
gi|9626243|ref|NC_001416.1| 14917 . CCGGCG CCG 228.409 .
INDEL;IDV=35;IMF=0.921053;DP=38;VDB=0.3497;SGB=-0.693136;RPBZ=1.29948;MQBZ=2.13501;MQSBZ=0.774597;BQBZ=-1.12422;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,0,21,14;MQ=40 GT:PL:AD 1/1:255,93,0:1,35
gi|9626243|ref|NC_001416.1| 15082 . CA C 202.363 .
INDEL;IDV=43;IMF=0.877551;DP=49;VDB=0.432542;SGB=-0.693146;RPBZ=-0.838871;MQBZ=-1.14318;MQSBZ=-3.11415;BQBZ=-1.33488;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,2,27,16;MQ=41 GT:PL:AD 1/1:229,106,0:2,43
gi|9626243|ref|NC_001416.1| 17658 . GA G 228.322 .
INDEL;IDV=44;IMF=0.93617;DP=47;VDB=0.414993;SGB=-0.693145;RPBZ=0.326509;MQBZ=-0.611046;MQSBZ=-0.63901;BQBZ=0.22239;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,2,24,17;MQ=41 GT:PL:AD 1/1:255,91,0:3,41
gi|9626243|ref|NC_001416.1| 19073 . GTGA G 228.358 .

INDEL;IDV=50;IMF=0.892857;DP=56;VDB=0.212885;SGB=-0.693147;RPBZ=-1.93428;MQBZ=-0.633879;MQSBZ=-0.186151;BQBZ=-1.72175;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,3,33,16;MQ=40 GT:PL:AD 1/1:255,105,0:4,49
gi|9626243|ref|NC_001416.1| 20591 . AGAGG AG 228.389 .
INDEL;IDV=55;IMF=0.916667;DP=60;VDB=0.123734;SGB=-0.693147;RPBZ=-0.187298;MQBZ=0.51719;MQSBZ=-1.45686;BQBZ=-0.72314;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=2,0,26,27;MQ=41 GT:PL:AD 1/1:255,134,0:2,53
gi|9626243|ref|NC_001416.1| 21054 . AACC A 228.318 .
INDEL;IDV=47;IMF=0.854545;DP=55;VDB=0.0751605;SGB=-0.693147;RPBZ=0.131374;MQBZ=0.746048;MQSBZ=0.302249;BQBZ=-1.05755;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,3,24,23;MQ=40 GT:PL:AD 1/1:255,97,0:4,47
gi|9626243|ref|NC_001416.1| 21533 . AG A 228.332 .
INDEL;IDV=40;IMF=0.952381;DP=42;VDB=0.0289263;SGB=-0.693145;RPBZ=-2.36336;MQBZ=-0.223607;MQSBZ=1.08465;BQBZ=-0.975412;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=2,0,17,23;MQ=41 GT:PL:AD 1/1:255,95,0:2,40
gi|9626243|ref|NC_001416.1| 21972 . AATGGCAT A 228.266 .
INDEL;IDV=34;IMF=0.871795;DP=39;VDB=0.00490472;SGB=-0.693132;RPBZ=-0.189191;MQBZ=0.58888;MQSBZ=-1.36883;BQBZ=-0.839881;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=3,1,19,15;MQ=36 GT:PL:AD 1/1:255,59,0:4,34
gi|9626243|ref|NC_001416.1| 22454 . CCTC CC 228.393 .
INDEL;IDV=27;IMF=0.931035;DP=29;VDB=0.264618;SGB=-0.692562;RPBZ=-1.89393;MQBZ=-1.44516;MQSBZ=-1.13748;BQBZ=-1.65995;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,0,9,13;MQ=40 GT:PL:AD 1/1:255,56,0:1,22
gi|9626243|ref|NC_001416.1| 22473 . GAA G 210.351 .
INDEL;IDV=25;IMF=0.892857;DP=28;VDB=0.315921;SGB=-0.692067;RPBZ=-0.928731;MQBZ=-0.0411597;MQSBZ=-0.848734;BQBZ=-1.66763;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,1,7,13;MQ=40 GT:PL:AD 1/1:237,43,0:2,20
gi|9626243|ref|NC_001416.1| 23112 . TAAAAAA TAAAA 225.417 .
INDEL;IDV=34;IMF=0.944444;DP=36;VDB=0.0452312;SGB=-0.693132;RPBZ=0.0691192;MQBZ=-0.5073;MQSBZ=-1.4276;BQBZ=-1.34761;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,0,15,19;MQ=41 GT:PL:AD 1/1:255,102,0:0,34
gi|9626243|ref|NC_001416.1| 23386 . TTAC T 225.417 .
INDEL;IDV=30;IMF=0.909091;DP=33;VDB=0.163818;SGB=-0.693097;RPBZ=2.81919;MQBZ=1.92558;MQSBZ=-0.170697;BQBZ=-1.75664;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,0,17,13;MQ=41 GT:PL:AD 1/1:255,90,0:0,30
gi|9626243|ref|NC_001416.1| 25184 . GTT GT 228.371 .
INDEL;IDV=37;IMF=0.948718;DP=39;VDB=0.556018;SGB=-0.693136;RPBZ=-2.22918;MQBZ=-0.232495;MQSBZ=0.816497;BQBZ=-0.249101;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,0,19,16;MQ=42 GT:PL:AD 1/1:255,93,0:1,35
gi|9626243|ref|NC_001416.1| 25920 . ATTT ATT 222.388 .
INDEL;IDV=38;IMF=0.95;DP=40;VDB=0.672712;SGB=-0.693141;RPBZ=2.17312;MQBZ=2.31176;MQSBZ=0.220246;BQBZ=-0.631446;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,1,15,22;MQ=41 GT:PL:AD 1/1:249,99,0:1,37
gi|9626243|ref|NC_001416.1| 26294 . CTATTATT CTATT 228.396 .
INDEL;IDV=34;IMF=0.918919;DP=37;VDB=0.50051;SGB=-0.693132;RPBZ=-0.974015;MQBZ=-0.321934;MQSBZ=-0.41831;BQBZ=0;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,1,20,14;MQ=40 GT:PL:AD 1/1:255,80,0:2,34
gi|9626243|ref|NC_001416.1| 27234 . GCTCT GCT 225.417 .
INDEL;IDV=28;IMF=0.965517;DP=29;VDB=0.0904072;SGB=-0.693054;RPBZ=-1.67476;MQBZ=-0.339683;MQSBZ=1.54638;BQBZ=-1.59098;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,0,18,10;MQ=41 GT:PL:AD 1/1:255,84,0:0,28
gi|9626243|ref|NC_001416.1| 27472 . GCACACA GCACA 228.35 .

INDEL;IDV=34;IMF=0.871795;DP=39;VDB=0.376907;SGB=-0.693132;RPBZ=-0.462467;MQBZ=0.94951;MQSBZ=-1.08616;BQBZ=-2.28301;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,3,15,19;MQ=41 GT:PL:AD 1/1:255,72,0:3,34

gi|9626243|ref|NC_001416.1| 30577 . CTTCTATTCT CTTCT 228.276 .

INDEL;IDV=39;IMF=0.764706;DP=51;VDB=0.0164896;SGB=-0.693144;RPBZ=-2.55488;MQBZ=-2.32421;MQSBZ=-0.968456;BQBZ=-0.953821;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=3,4,18,21;MQ=39 GT:PL:AD 1/1:255,60,0:7,39

gi|9626243|ref|NC_001416.1| 33181 . AGGGGG AGGG 228.339 .

INDEL;IDV=39;IMF=0.866667;DP=45;VDB=0.169757;SGB=-0.693144;RPBZ=1.70368;MQBZ=-0.812658;MQSBZ=1.43486;BQBZ=0.578169;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=2,1,19,20;MQ=41 GT:PL:AD 1/1:255,85,0:3,39

gi|9626243|ref|NC_001416.1| 34442 . TA T 193.293 .

INDEL;IDV=33;IMF=0.891892;DP=37;VDB=0.219282;SGB=-0.693127;RPBZ=-2.66695;MQBZ=3.25507;MQSBZ=-1.0328;BQBZ=-0.504286;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,2,15,18;MQ=41 GT:PL:AD 1/1:220,70,0:3,33

gi|9626243|ref|NC_001416.1| 36449 . AT A 228.362 .

INDEL;IDV=38;IMF=0.926829;DP=41;VDB=0.0615883;SGB=-0.693143;RPBZ=0.951563;MQBZ=-0.661371;MQSBZ=0.0616041;BQBZ=-0.366355;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,2,19,19;MQ=41 GT:PL:AD 1/1:255,91,0:2,38

gi|9626243|ref|NC_001416.1| 36957 . GTTTTT GTTT 228.282 .

INDEL;IDV=29;IMF=0.828571;DP=35;VDB=0.00810286;SGB=-0.693079;RPBZ=-1.05097;MQBZ=0.661699;MQSBZ=1.67168;BQBZ=-0.128447;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,2,12,17;MQ=41 GT:PL:AD 1/1:255,50,0:3,29

gi|9626243|ref|NC_001416.1| 38381 . AAC A 228.381 .

INDEL;IDV=37;IMF=0.880952;DP=42;VDB=0.283513;SGB=-0.693141;RPBZ=-0.641107;MQBZ=-0.763685;MQSBZ=-1.36583;BQBZ=-0.754591;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=3,0,12,25;MQ=41 GT:PL:AD 1/1:255,80,0:3,37

gi|9626243|ref|NC_001416.1| 39006 . ACC AC 208.206 .

INDEL;IDV=30;IMF=0.833333;DP=36;VDB=0.308515;SGB=-0.693097;RPBZ=-2.27168;MQBZ=-0.185695;MQSBZ=-1.05529;BQBZ=0.402448;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=3,0,11,19;MQ=41 GT:PL:AD 1/1:235,57,0:3,30

gi|9626243|ref|NC_001416.1| 39900 . GTTGCAT GT 228.404 .

INDEL;IDV=34;IMF=0.944444;DP=36;VDB=0.0309817;SGB=-0.693132;RPBZ=0;MQBZ=-1.14462;MQSBZ=-0.959235;BQBZ=-0.437332;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,1,18,16;MQ=41 GT:PL:AD 1/1:255,90,0:1,34

gi|9626243|ref|NC_001416.1| 40831 . AT A 225.417 .

INDEL;IDV=28;IMF=1;DP=28;VDB=0.184255;SGB=-0.693054;MQSBZ=0;BQBZ=-1.2684;MQ0F=0;AC=2;AN=2;DP4=0,0,12,16;MQ=42 GT:PL:AD 1/1:255,84,0:0,28

gi|9626243|ref|NC_001416.1| 41193 . GTTGT G 228.308 .

INDEL;IDV=37;IMF=0.925;DP=40;VDB=0.00133078;SGB=-0.69168;RPBZ=0.796405;MQBZ=1.54685;MQSBZ=2.24984;BQBZ=-1.57129;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=2,0,10,9;MQ=33 GT:PL:AD 1/1:255,38,0:2,19

gi|9626243|ref|NC_001416.1| 41253 . CGCTACCGTATTG C 136.176 .

INDEL;IDV=27;IMF=0.818182;DP=33;VDB=0.0263282;SGB=-0.692562;RPBZ=-0.793625;MQBZ=-1.78062;MQSBZ=0.649234;BQBZ=-1.23304;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,2,15,7;MQ=19 GT:PL:AD 1/1:163,39,0:3,22

gi|9626243|ref|NC_001416.1| 41678 . CTT C 228.41 .

INDEL;IDV=50;IMF=0.980392;DP=51;VDB=0.684289;SGB=-0.693147;RPBZ=-1.5293;MQBZ=2.10489;MQSBZ=-0.0591312;BQBZ=-0.551736;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,0,20,30;MQ=41 GT:PL:AD 1/1:255,136,0:1,50

gi|9626243|ref|NC_001416.1| 42691 . CCGT C 228.412 .

INDEL;IDV=34;IMF=0.971429;DP=35;VDB=0.074891;SGB=-0.693127;RPBZ=-1.68419;MQBZ=-1

```
.1024;MQSBZ=0.255476;BQBZ=-1.63615;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=0,1,11,22;MQ=39
GT:PL:AD    1/1:255,87,0:1,33
gi|9626243|ref|NC_001416.1|    42798 .    GCACC GC    228.403 .
INDEL;IDV=44;IMF=0.897959;DP=49;VDB=0.0646112;SGB=-0.693145;RPBZ=-1.7512;MQBZ=-0
.394671;MQSBZ=-1.72785;BQBZ=-1.51562;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=0,2,21,20;MQ=4
0 GT:PL:AD    1/1:255,100,0:2,41
gi|9626243|ref|NC_001416.1|    44172 .    TGCA T    228.367 .
INDEL;IDV=42;IMF=0.893617;DP=47;VDB=0.0680788;SGB=-0.693145;RPBZ=0.724904;MQBZ=
0.144255;MQSBZ=0.422803;BQBZ=-1.69283;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=2,2,21,20;MQ
=41 GT:PL:AD    1/1:255,84,0:4,41
gi|9626243|ref|NC_001416.1|    44335 .    CGG CG    225.417 .
INDEL;IDV=42;IMF=0.893617;DP=47;VDB=0.84515;SGB=-0.693146;RPBZ=0.949142;MQBZ=-0.
40071;MQSBZ=-1.0774;BQBZ=1.00443;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=0,0,20,22;MQ=41
GT:PL:AD    1/1:255,126,0:0,42
gi|9626243|ref|NC_001416.1|    45329 .    AACGAT A    228.298 .
INDEL;IDV=31;IMF=0.885714;DP=35;VDB=0.0538266;SGB=-0.69311;RPBZ=-1.53047;MQBZ=-1
.19372;MQSBZ=0.305503;BQBZ=-0.750114;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=2,1,13,18;MQ=
38 GT:PL:AD    1/1:255,59,0:3,31
gi|9626243|ref|NC_001416.1|    45558 .    TAA T    228.358 .
INDEL;IDV=39;IMF=0.975;DP=40;VDB=0.445349;SGB=-0.693144;RPBZ=-1.68959;MQBZ=-0.76
4719;MQSBZ=-0.699548;BQBZ=1.26194;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=0,1,21,18;MQ=40
GT:PL:AD    1/1:255,100,0:1,39
gi|9626243|ref|NC_001416.1|    45887 .    ATG A    228.28 .
INDEL;IDV=46;IMF=0.901961;DP=51;VDB=0.227666;SGB=-0.693147;RPBZ=-3.64327;MQBZ=-0
.850331;MQSBZ=0.39101;BQBZ=-1.41179;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=3,1,18,28;MQ=4
1 GT:PL:AD    1/1:255,88,0:4,46
```

```
grep -v '^#' tmp.vcf | grep 'INDEL' | awk '{if ($4 < $5) print $0}'
# Insertion
```

结果为

```
gi|9626243|ref|NC_001416.1|    363 .    A    ATG    35.1841 .
INDEL;IDV=8;IMF=0.228571;DP=35;VDB=0.00843033;SGB=-0.636426;RPBZ=-3.32157;MQBZ=
2.68634;MQSBZ=-2.15744;BQBZ=0.587995;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=7,13,4,3;MQ=2
8 GT:PL:AD    1/1:63,6,0:20,7
```

4 统计 INDEL 条目的突变组合分布频率

```
grep -v '^#' tmp.vcf | grep 'INDEL' | cut -f 4-5 | tr '\t' '-' | sort | uniq -c
```

结果为

```
1 A-ATG
1 AAC-A
1 AACC-A
1 AACGAT-A
1 AATGGCAT-A
1 ACC-AC
```

1 ACGC-AC
2 AG-A
1 AGAGG-AG
1 AGCAGG-AG
1 AGGATG-AG
1 AGGGGG-AGGG
2 AT-A
1 ATG-A
1 ATT-AT
1 ATTT-ATT
2 CA-C
1 CAGCT-C
1 CCCACC-CCC
1 CCGGCG-CCG
1 CCGT-C
1 CCGTGTGGCAGAGCTGAA-C
1 CCTC-CC
1 CGCTCACCGTATTG-C
1 CGG-CG
1 CTATTATT-CTATT
1 CTT-C
1 CTTCTATTCT-CTTCT
2 GA-G
1 GAA-G
1 GCACACA-GCACA
1 GCACC-GC
1 GCTCT-GCT
1 GCTGAAATTGA-G
1 GT-G
1 GTGA-G
1 GTT-GT
1 GTTGCAT-GT
1 GTTGT-G
1 GTTTTT-GTTT
1 TA-T
1 TAA-T
1 TAAAAA-TAAAA
1 TCACCC-TC
1 TCC-TC
1 TCCACGCC-TCC
1 TGC-T
1 TGCA-T
1 TTAC-T

5 找到基因型不是 1/1 的条目

```
grep -v '^#' tmp.vcf | awk '{if ($10 !~ "1/1") print}'
```

结果为

(none)

6 筛选测序深度大于 30 的条目

```
egrep 'DP=3[1-9]|DP=[4-9][0-9]' tmp.vcf | less -N
```

结果为

```
1 gi|9626243|ref|NC_001416.1| 245 . ATT AT 216.329 .  
INDEL;IDV=33;IMF=0.942857;DP=35;VDB=0.537812;SGB=-0.69311;RPBZ=-1.49293;MQBZ=1.5  
8942;MQSBZ=2.4775  
1 9;BQBZ=-1.57202;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=1,0,16,15;MQ=36 GT:PL:AD  
1/1:243,79,0:1,31  
2 gi|9626243|ref|NC_001416.1| 353 . GCTGAAATTGA G 222.98 .  
INDEL;IDV=24;IMF=0.685714;DP=35;VDB=0.00608521;SGB=-0.692831;RPBZ=1.47495;MQBZ=  
-2.92683;M  
2 QSBZ=1.11022;BQBZ=-1.80962;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=5,5,12,12;MQ=28  
GT:PL:AD 1/1:250,9,0:10,24  
3 gi|9626243|ref|NC_001416.1| 363 . A ATG 35.1841 .  
INDEL;IDV=8;IMF=0.228571;DP=35;VDB=0.00843033;SGB=-0.636426;RPBZ=-3.32157;MQBZ=  
2.68634;MQSBZ=-2.1  
3 5744;BQBZ=0.587995;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=7,13,4,3;MQ=28 GT:PL:AD  
1/1:63,6,0:20,7  
4 gi|9626243|ref|NC_001416.1| 1104 . C A 228.378 .  
DP=43;VDB=0.337802;SGB=-0.693141;RPBZ=-1.32279;MQBZ=-0.164399;MQSBZ=-0.638285;B  
QBZ=1.04957;SCBZ=0 4;MQOF=0;AC=2;AN=2;DP4=1,0,10,27;MQ=41 GT:PL:AD  
1/1:255,96,0:1,36  
5 gi|9626243|ref|NC_001416.1| 1344 . G T 222.417 .  
DP=37;VDB=0.730004;SGB=-0.693097;MQSBZ=0.816497;MQOF=0;AC=2;AN=2;DP4=0,0,18,12;  
MQ=41 GT:PL:AD 5 1/1:252,87,0:0,29  
6 gi|9626243|ref|NC_001416.1| 2143 . C G 225.417 .  
DP=46;VDB=0.942003;SGB=-0.693141;MQSBZ=0;MQOF=0;AC=2;AN=2;DP4=0,0,16,21;MQ=42  
GT:PL:AD 1 6 /1:255,108,0:0,36  
7 gi|9626243|ref|NC_001416.1| 2817 . GA G 228.24 .  
INDEL;IDV=41;IMF=0.911111;DP=45;VDB=0.27095;SGB=-0.693144;RPBZ=-0.797953;MQBZ=-1  
.17465;MQSBZ=1.36 7  
281;BQBZ=-0.609359;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=2,1,20,19;MQ=41 GT:PL:AD  
1/1:255,80,0:3,39  
8 gi|9626243|ref|NC_001416.1| 2952 . CCCACC CCC 228.295 .  
INDEL;IDV=28;IMF=0.903226;DP=31;VDB=0.17228;SGB=-0.692914;RPBZ=-2.27241;MQBZ=-0.  
907724;MQSBZ=-0.8 8  
23524;BQBZ=-0.0964065;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=0,3,13,12;MQ=39 GT:PL:AD  
1/1:255,47,0:3,25  
9 gi|9626243|ref|NC_001416.1| 3264 . CA C 212.288 .  
INDEL;IDV=41;IMF=0.803922;DP=51;VDB=0.063464;SGB=-0.693145;RPBZ=1.97114;MQBZ=-1.  
39289;MQSBZ=0.302 9  
321;BQBZ=0.842334;SCBZ=0;MQOF=0;AC=2;AN=2;DP4=2,2,18,23;MQ=41 GT:PL:AD  
1/1:239,84,0:4,41  
10 gi|9626243|ref|NC_001416.1| 3316 . T C 228.39 .  
DP=59;VDB=0.4619;SGB=-0.693146;RPBZ=1.00132;MQBZ=-0.353553;MQSBZ=1.35534;BQBZ  
=1.07902;SCBZ=0;MQOF 10 =0;AC=2;AN=2;DP4=0,1,23,21;MQ=41 GT:PL:AD  
1/1:255,116,0:1,43
```


7 筛选变异位点质量值大于 40 的条目

```
grep -v '^#' tmp.vcf | awk '{if ($6 > 30) print}' | less -N
```

结果为

```
1 gi|9626243|ref|NC_001416.1| 245 . ATT AT 216.329 .
INDEL;IDV=33;IMF=0.942857;DP=35;VDB=0.537812;SGB=-0.69311;RPBZ=-1.49293;MQBZ=1.5
8942;MQSBZ=2.4775
1 9;BQBZ=-1.57202;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,0,16,15;MQ=36 GT:PL:AD
1/1:243,79,0:1,31
2 gi|9626243|ref|NC_001416.1| 353 . GCTGAAATTGA G 222.98 .
INDEL;IDV=24;IMF=0.685714;DP=35;VDB=0.00608521;SGB=-0.692831;RPBZ=1.47495;MQBZ=
-2.92683;M
2 QSBZ=1.11022;BQBZ=-1.80962;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=5,5,12,12;MQ=28
GT:PL:AD 1/1:250,9,0:10,24
3 gi|9626243|ref|NC_001416.1| 363 . A ATG 35.1841 .
INDEL;IDV=8;IMF=0.228571;DP=35;VDB=0.00843033;SGB=-0.636426;RPBZ=-3.32157;MQBZ=
2.68634;MQSBZ=-2.1 3
5744;BQBZ=0.587995;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=7,13,4,3;MQ=28 GT:PL:AD
1/1:63,6,0:20,7
4 gi|9626243|ref|NC_001416.1| 1104 . C A 228.378 .
DP=43;VDB=0.337802;SGB=-0.693141;RPBZ=-1.32279;MQBZ=-0.164399;MQSBZ=-0.638285;B
QBZ=1.04957;SCBZ=0 4;MQ0F=0;AC=2;AN=2;DP4=1,0,10,27;MQ=41 GT:PL:AD
1/1:255,96,0:1,36
5 gi|9626243|ref|NC_001416.1| 1344 . G T 222.417 .
DP=37;VDB=0.730004;SGB=-0.693097;MQSBZ=0.816497;MQ0F=0;AC=2;AN=2;DP4=0,0,18,12;
MQ=41 GT:PL:AD 5 1/1:252,87,0:0,29
6 gi|9626243|ref|NC_001416.1| 2143 . C G 225.417 .
DP=46;VDB=0.942003;SGB=-0.693141;MQSBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,0,16,21;MQ=42
GT:PL:AD 1 6 /1:255,108,0:0,36
7 gi|9626243|ref|NC_001416.1| 2817 . GA G 228.24 .
INDEL;IDV=41;IMF=0.911111;DP=45;VDB=0.27095;SGB=-0.693144;RPBZ=-0.797953;MQBZ=-1
.17465;MQSBZ=1.36 7
281;BQBZ=-0.609359;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=2,1,20,19;MQ=41 GT:PL:AD
1/1:255,80,0:3,39
8 gi|9626243|ref|NC_001416.1| 2952 . CCCACC CCC 228.295 .
INDEL;IDV=28;IMF=0.903226;DP=31;VDB=0.17228;SGB=-0.692914;RPBZ=-2.27241;MQBZ=-0.
907724;MQSBZ=-0.8 8
23524;BQBZ=-0.0964065;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,3,13,12;MQ=39 GT:PL:AD
1/1:255,47,0:3,25
9 gi|9626243|ref|NC_001416.1| 3264 . CA C 212.288 .
INDEL;IDV=41;IMF=0.803922;DP=51;VDB=0.063464;SGB=-0.693145;RPBZ=1.97114;MQBZ=-1.
39289;MQSBZ=0.302 9
321;BQBZ=0.842334;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=2,2,18,23;MQ=41 GT:PL:AD
1/1:239,84,0:4,41
10 gi|9626243|ref|NC_001416.1| 3316 . T C 228.39 .
DP=59;VDB=0.4619;SGB=-0.693146;RPBZ=1.00132;MQBZ=-0.353553;MQSBZ=1.35534;BQBZ
=1.07902;SCBZ=0;MQ0F 10 =0;AC=2;AN=2;DP4=0,1,23,21;MQ=41 GT:PL:AD
1/1:255,116,0:1,43
```

8 组合筛选变异位点质量值大于 30 并且深度大于 20 的条目

```
grep -v '^#' tmp.vcf | awk '{if ($6 > 30) print}' | egrep 'DP=2[1-9]|DP=[3-9][0-9]' | less -N
```

结果为

```
1 gi|9626243|ref|NC_001416.1| 245 . ATT AT 216.329 .  
INDEL;IDV=33;IMF=0.942857;DP=35;VDB=0.537812;SGB=-0.69311;RPBZ=-1.49293;MQBZ=1.5  
8942;MQSBZ=2.4775  
1 9;BQBZ=-1.57202;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=1,0,16,15;MQ=36 GT:PL:AD  
1/1:243,79,0:1,31  
2 gi|9626243|ref|NC_001416.1| 353 . GCTGAAATTGA G 222.98 .  
INDEL;IDV=24;IMF=0.685714;DP=35;VDB=0.00608521;SGB=-0.692831;RPBZ=1.47495;MQBZ=  
-2.92683;M  
2 QSBZ=1.11022;BQBZ=-1.80962;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=5,5,12,12;MQ=28  
GT:PL:AD 1/1:250,9,0:10,24  
3 gi|9626243|ref|NC_001416.1| 363 . A ATG 35.1841 .  
INDEL;IDV=8;IMF=0.228571;DP=35;VDB=0.00843033;SGB=-0.636426;RPBZ=-3.32157;MQBZ=  
2.68634;MQSBZ=-2.1  
3 5744;BQBZ=0.587995;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=7,13,4,3;MQ=28 GT:PL:AD  
1/1:63,6,0:20,7  
4 gi|9626243|ref|NC_001416.1| 1104 . C A 228.378 .  
DP=43;VDB=0.337802;SGB=-0.693141;RPBZ=-1.32279;MQBZ=-0.164399;MQSBZ=-0.638285;B  
QBZ=1.04957;SCBZ=0  
4;MQ0F=0;AC=2;AN=2;DP4=1,0,10,27;MQ=41 GT:PL:AD 1/1:255,96,0:1,36  
5 gi|9626243|ref|NC_001416.1| 1344 . G T 222.417 .  
DP=37;VDB=0.730004;SGB=-0.693097;MQSBZ=0.816497;MQ0F=0;AC=2;AN=2;DP4=0,0,18,12;  
MQ=41 GT:PL:AD  
5 1/1:252,87,0:0,29  
6 gi|9626243|ref|NC_001416.1| 2143 . C G 225.417 .  
DP=46;VDB=0.942003;SGB=-0.693141;MQSBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,0,16,21;MQ=42  
GT:PL:AD 1  
6 /1:255,108,0:0,36  
7 gi|9626243|ref|NC_001416.1| 2817 . GA G 228.24 .  
INDEL;IDV=41;IMF=0.911111;DP=45;VDB=0.27095;SGB=-0.693144;RPBZ=-0.797953;MQBZ=-1  
.17465;MQSBZ=1.36  
7 281;BQBZ=-0.609359;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=2,1,20,19;MQ=41 GT:PL:AD  
1/1:255,80,0:3,39  
8 gi|9626243|ref|NC_001416.1| 2952 . CCCACC CCC 228.295 .  
INDEL;IDV=28;IMF=0.903226;DP=31;VDB=0.17228;SGB=-0.692914;RPBZ=-2.27241;MQBZ=-0.  
907724;MQSBZ=-0.8  
8 23524;BQBZ=-0.0964065;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=0,3,13,12;MQ=39 GT:PL:AD  
1/1:255,47,0:3,25  
9 gi|9626243|ref|NC_001416.1| 3264 . CA C 212.288 .  
INDEL;IDV=41;IMF=0.803922;DP=51;VDB=0.063464;SGB=-0.693145;RPBZ=1.97114;MQBZ=-1.  
39289;MQSBZ=0.302  
9 321;BQBZ=0.842334;SCBZ=0;MQ0F=0;AC=2;AN=2;DP4=2,2,18,23;MQ=41 GT:PL:AD  
1/1:239,84,0:4,41  
10 gi|9626243|ref|NC_001416.1| 3316 . T C 228.39 .
```

DP=59;VDB=0.4619;SGB=-0.693146;RPBZ=1.00132;MQBZ=-0.353553;MQSBZ=1.35534;BQBZ=1.07902;SCBZ=0;MQ0F
10 =0;AC=2;AN=2;DP4=0,1,23,21;MQ=41 GT:PL:AD 1/1:255,116,0:1,43

9 计算每个变异位点的等位基因频率

```
grep -v "^#" tmp.vcf | cut -f 8 | egrep -o '[0-9]+,[0-9]+,[0-9]+,[0-9]+' |  
egrep -o '[0-9]+,[0-9]+,[0-9]+,[0-9]+' | awk -v FS=","  
'{print($3+$4)/($1+$2+$3+$4)}'
```

结果为

0.96875
0.705882
1
1
0.259259
0.973684
1
1
0.928571
0.892857
0.911111
0.977778
1
0.925926
0.857143
0.974359
1
1
1
0.888889
1
1
0.821429
0.916667
1
0.861111
0.945946
0.869565
0.948718
1
0.972222
0.955556
0.931818
1
0.924528
0.963636
0.921569
1
0.972973

0.952381
1
1
0.894737
0.956522
0.909091
1
1
1
1
0.972222
0.964286
0.973684
0.944444
1
1
0.918919
1
0.847826
1
0.928571
0.916667
1
1
0.95
0.90625
1
1
0.925
0.909091
1
0.971429
1
1
1
0.904762
0.88
1
0.980392
0.970588
0.953488
1
0.911111
1
0.911765
0.975
1
0.92
1
1
1

10 从 vcf 文件挑选任意一个突变位点的测序深度表明的那些 reads

选第33行

```
gi|9626243|ref|NC_001416.1| 360 . T G 8.13869 .  
DP=13;VDB=0.00637373;SGB=-0.636426;MQSBZ=-1.77482;MQOF=0;AC=2;AN=2;DP4=0,0,4,3;  
MQ=31 GT:PL:AD 1/1:37,21,0:0,7
```

```
samtools view tmp.bam | awk '{if ($6 != "*" && $4 <= 1104 && substr($10,  
360-$4+1, 1) == "G") print}' | less -N
```

结果为

```
1 r6698 147 gi|9626243|ref|NC_001416.1| 343 24 21M2I60M = 130 -294  
AAAATGGTATGCCCAAAGGGATGGAACGAAAAGCTGCGCCGGGAGGTTGAAGAACTGCGNCAGGC  
CAGCGAGG  
  
1 CAGATCTCCA )#%('#!&!+*(&%#&""'%$'%"**#+++)("+#$#+"+"#$(("&*("$%  
(!#'+')"$()#%'"+&"&!+($! AS:i:-28 XN:i:0 XM:i:9 XO:i:1 XG:i:2 NM:i:11  
MD:Z:0C0G2A2G4T0G  
  
1 3T0T38G23 YS:i:0 YT:Z:CP  
  
2 r9595 83 gi|9626243|ref|NC_001416.1| 345 24 19M2I99M = 252 -211  
AATGGTATGCCGAAAGGGATGGAACGAAAAGCTGCGCCGGGAGGTTGAAGAACTGCGGCAGGCCA  
GCGAGGCA  
  
2 GATCTCCAGCCAGGAAGTATTGAGTACGAACGCCATCGACTTACGCG >51$+10FD"C4435&-  
FF<,B70C5H=E8%(!/B7#7)/7<H!50CG4A*+(%2>,"3+2#;F2%4,6!541H>B8*0)6=H!%-  
<)C4.9=E+96<66%+4(DG6G7!F:#E1/5- A  
  
2 S:i:-25 XN:i:0 XM:i:5 XO:i:1 XG:i:2 NM:i:7 MD:Z:2A2G4T4T0T101 YS:i:-36 YT:Z:CP  
  
3 r7839 163 gi|9626243|ref|NC_001416.1| 348 24 16M2I44M = 563 309  
GGTATGCCGAAAGGGATGGAACGAAAAGCTGCGCCGGGAGGTTGAAGAACTGCGGCAGGCCA  
,B'.2$?E'  
  
3 ?GF6FF14)H4.E#++?!)DH5/C"9)H8+$$;!4@A<B$A0>,+<1=*$;>5 AS:i:-27 XN:i:0 XM:i:4  
XO:i:1 XG:i:2 NM:i:6 MD:Z:2G4T4T0T46 YS:i:-2 YT:Z:CP  
  
4 r8654 147 gi|9626243|ref|NC_001416.1| 349 40 15M2I42M = 112 -294  
GTATGCCNAAAGGGANGGAACGAAAAGCTGCGCCGGGAGGTTGAAGAACTGCGGNAGGC  
'#$*"&&'<br>4 %!"%*"")(#*+(&""!)$'%*(+""+*&)%!(!!**&'!+#!)$() AS:i:-21 XN:i:0 XM:i:6 XO:i:1  
XG:i:2 NM:i:8 MD:Z:1G4T0G3T0T39C4 YS:i:-9 YT:Z:CP  
  
5 r510 163 gi|9626243|ref|NC_001416.1| 350 40 14M2I58M = 409 192  
TANNNCGAAAGGGATGGAACGAAAAGCTGCGCCGGGAGGTTGAAGAACTGCGGCAGGCCAGCGA  
GGCAGATCT  
  
5 C '%!&%)**+*#"(!((##!)*&*$#&"&#!%*+(*$%&$)&(("%*+!#+'$+(&#%*(%%%$!+"&%(&  
AS:i:-22 XN:i:0 XM:i:7 XO:i:1 XG:i:2 NM:i:9 MD:Z:0G1T0G0C0T4T0T60 YS:i:-3 Y  
  
5 T:Z:CP
```

6 r8425 99 gi|9626243|ref|NC_001416.1| 353 42 11M2I295M = 476 306
GCCGAAAGGGATGGAACGAAAAGCTGCGCCGGGAGGTTGAAGAACTGCGGCAGGCCAGCGAGGC
AGATCTCCA

6
GCCAGGAACTATTGAGTACGAACGCCATCGACTTACGCGTGCGCAGGCCGACGCACAGGAACTGAA
GAATGCCAGAGACTCCGCTGNAGTGGTGGAAACCGCATTCTGTACTTTCGTGCTGTCGCGGATCGC
AGGTGAAATTGCCAGTATTCTCNACGGGCTCCCCCTGTCGGTGCA

6 GCGGCGTTTTCGGGAAGTGGAAAACCGACATGTTGATTTCTGAAACGGGATATCATC ,2-
6.86\$BAD;((B=-%H3!+2?H2"#5D8)E)/G2>*?,"=:65DE?!5.?1)"!#F9?63-9C>8+"(:5%2CA@7??
G'>/5F9%73."1F-BE*E<74<#:@46+7B1

6 0=(\$77\$4H>\$"H0+A>;AE.B%"7(\$HB0%-->'*"++;#,-!CC9H0283E(*-/2=<G=>*;;9(!
<8"GC,A(66<5B?.67A#3A\$9E8=B<@DE+&=308F>4C8'%%+%":!,45/7#,1<GD/H;03)G<!/&>--
AB%/D0&48"8\$B/*";9729519;>:D@<

6 1%<3-\$D6C<.!);/0, AS:i:-28 XN:i:0 XM:i:6 XO:i:1 XG:i:2 NM:i:8
MD:Z:2T4T0T148A67G33C46 YS:i:-1 YT:Z:CP

7 r3988 163 gi|9626243|ref|NC_001416.1| 354 42 10M2I257M = 560 316
ACGAAAGGGATGGAACGAAAAGCTGCGCCGGGAGGTTGAAGAACTGCGGCAGGGCAGCGAGGCA
GATCTCCAG

7
CCNGGAACTATTGAGTACGAACGCCATCGACTTACGCGTGCGCAGGCCGACGCACAGGAACTGAAG
AATGCCAGAGACTCCGNTGAAGTGGNGGANACCGCATTCTGTACTTTCGTGCTGTCGCGGATAGCA
GGTGAATNGCCAGTATTCTCGACGGGNTCCCCCTGTCGGTGCA

7 CGGCGTTTTCCNGAACTGG %'5227)G52&+B%91?/#'7---FB)*.;(24;!<C"!87-2;%A2"F!=@2A.>
<2A2)F'@5/,91(A)91FF1*6):?H"3-!08,/=FG+,%AHH)D\$+<("84D%\$#1B?
BD1%06H8133D,)/5;>>0G7!%5C-%F>9.8B

7 2\$2D29&.1*%@;,#0+?>7@(1!=AH?593;H.7?9-E/D4H4!+B)9;.0;9=>1D/ED1+A-8&<16@(7?
-8="#G3/%A7"2G35>-)#B"=C.%-\$>1D*&.C-.554\$F AS:i:-38 XN:i:0 XM:i:13 XO:i:1 XG:i:2
NM:i:15 M

7 D:Z:0C0T4T0T43C21A79C8T3A32C11T18C28G7 YS:i:-2 YT:Z:CP

8 r9990 83 gi|9626243|ref|NC_001416.1| 357 23 41M = 208 -190
AGGGATGGAACGAGAAGCTGCGCCGGGAGGTTGAAGAACTG 0/3\$\$6-B\$#8%))/;>&&\$0(4?
79@4(-#38

8 GF\$218?8 AS:i:-19 XN:i:0 XM:i:7 XO:i:0 XG:i:0 NM:i:7 MD:Z:1A0A0T0T0G0A6A27
YS:i:-43 YT:Z:CP

9 r5552 147 gi|9626243|ref|NC_001416.1| 359 23 62M = 239 -182
GGATGGAACGAAAAGCTGCGCCGGGAGGTTGNAGAACTGCGGCAGGCCCGCGANGCAGNTCT
!"*)*&('+\$+%#!")!

9 +'\$!'+#\$(")""())#'*\$#'%\$)&\$+(*!#*(&!")##*)* AS:i:-15 XN:i:0 XM:i:9 XO:i:0 XG:i:0
NM:i:9 MD:Z:0A0T0T0G0A26A16A4G4A3 YS:i:-44 YT:Z:CP

10 r2505 147 gi|9626243|ref|NC_001416.1| 360 40 93M = 135 -318
GATGGAACGAAAAGCTGCGCCGGGAGGTTGAAGAACTGCGGCAGGCCAGCGAGGCAGATCTCCAG
CCCCGTACTATTGNNN


```
10 NCGNACNNCINN #&!%++!!$+$""$!+!*#&"*#$""!!)!&'!+))!(!)*+"#"#$)%(!&&
(")""#&&!*!+%#(#))*&(!*!!)*&%!!)!"( AS:i:-23 XN:i:0 XM:i:16 XO:i:0 XG:i:0 NM:i:16
MD:Z:OTOT

10 OG0A63A2A7A0G0T0A2A2G0C1A0T0C0 YS:i:-5 YT:Z:CP
```

11 思考题

VCF 文件以##开头的是什么信息? '#'开头的是什么信息?

以##开头的信息是meta data \ head, 里面是描述文档的一些信息, 如VCF文件的版本、参考基因组的信息、格式定义等

以#开头的就一行, 这个是这张表的属性的属性名

VCF 文件的 ref, alt 列和样本列的 0/1,1/1,1/2 有什么联系?

0/0表示样本中该位点为纯合位点, 和REF的碱基类型一致

0/1表示样本中该位点为杂合突变, 有REF和ALT两个基因型 (部分碱基和REF碱基类型一致, 部分碱基和ALT碱基类型一致)

1/1表示样本中该位点为纯合突变, 总体突变类型和ALT碱基类型一致

1/2表示样本中该位点为杂合突变, 有ALT1和ALT2两个基因型 (部分和ALT1碱基类型一致, 部分和ALT2碱基类型一致)

VCF 文件第三列出现的 rs 号的 id 是什么?

变异位点名称, 对应dbSNP数据库中的ID